Supplementary material for "No Free Credit: How Much Simulated Evidence Can Replace Real Events in Zero-Failure Reliability Demonstration of Perception Systems"

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Contents: S1 proofs of C1–C5a with numerical checks; S2 units and event counts; S3 the full operating-point grid; S4 audit details and must-fail scenarios; S5 sensitivity analyses; S6 twin construction and asset licences; S7 all supplementary tables. All numbers are generated from the study's result files; Monte-Carlo checks use seed 42.

S1 Proofs of C1–C5a

Numbering in this section is local. Correspondence with the main text: Theorem S1 proves Result 1, Theorem S2 Result 2, Lemma S3 and Proposition S9 Result 3, Theorem S6, Proposition S7 and Proposition S8 Result 4(a), (b) and (c). The main text states the same results in its Appendix A.

S1.1 Setting and notation

Events and misses. A camera (scene) s produces events $i = 1, 2, \ldots$ (a maximal run of frames containing the target object). The system samples frames periodically with period K and an unknown phase $\varphi \in \{0, \ldots, K-1\}$; a detector is run on the sampled frames. An event is *missed* ($M = 1$) when no sampled frame of the event is a hit (the detector finds the object). With $H \subseteq \mathbb{Z}$ the hit frames of the event and φ uniform, $\mathbb{P}(M = 1 \mid H) = 1 - |H \bmod K|/K$, where $H \bmod K$ is the set of residues of H. The miss rate of camera s is $p_s = \mathbb{P}(M = 1)$ over its events and phases. Target: certify $p_s \le \varepsilon$ with confidence $1 - \delta$.

Clopper–Pearson bound. For $x \in \{0, \ldots, n\}$, $F_{n,p}(x) = \mathbb{P}(\mathrm{Bin}(n, p) \le x)$ and $U(x, n) = U_\delta(x, n)$ is the one-sided $(1 - \delta)$ upper Clopper–Pearson bound [1]: the solution u of $F_{n,u}(x) = \delta$ for $x < n$, and 1 for $x = n$. Since $F_{n,p}(x)$ is strictly decreasing in p, $U(x, n) < p \iff F_{n,p}(x) < \delta$, and U is non-decreasing in x. Validity: $\mathbb{P}(U(\mathrm{Bin}(n, p), n) < p) \le \delta$ for every p.

Success run. $n_0(\varepsilon, \delta) = \lceil \ln \delta / \ln(1 - \varepsilon) \rceil$ is the smallest n with $(1 - \varepsilon)^n \le \delta$ (e.g. $n_0(5\%, 5\%) = 59$, $n_0(20\%, 5\%) = 14$).

Simulator and twin. Simulator data Y have a law Q on an arbitrary space. A *twin* is a paired simulator: every real event i has a synthetic counterpart with miss indicator M_i' under the same sampling phase. $q = \mathbb{P}(M' = 1)$, and the *discordance* is $D = \mathbb{P}(M = 1, M' = 0)$ (real miss, twin catch).

S1.2 C1: no free credit

A certification procedure is a (possibly randomised) test $\phi(X, Y) \in \{0, 1\}$ with real data $X = (M_1, \ldots, M_n) \sim \mathrm{Bern}(p)^{\otimes n}$ independent of simulator data $Y \sim Q$. It is *valid at level δ on a class $\mathcal{P}$* of pairs (p, Q) if $\sup_{(p,Q)\in\mathcal{P},\, p>\varepsilon} \mathbb{P}_{p,Q}(\phi = 1) \le \delta$.

Theorem S1 (C1)**.** *Assume $\mathcal{P}$ places no restriction linking Q and p: for every Q there are $(0, Q) \in \mathcal{P}$ and $(p, Q) \in \mathcal{P}$ for all $p \in (\varepsilon, \varepsilon + \eta)$ for some $\eta > 0$. If ϕ is valid at level δ on $\mathcal{P}$, then for every Q and every simulator sample size,*

$$(1 - \varepsilon)^n \, \mathbb{P}_{0,Q}(\phi = 1) \le \delta.$$

In particular, certifying a perfect system ($p = 0$) with probability one requires $n \ge n_0(\varepsilon, \delta)$ real events, whatever the amount of simulator data; power $1 - \beta$ at $p = 0$ requires $n \ge \ln(\delta/(1 - \beta))/\ln(1 - \varepsilon)$.

Proof. Fix Q and write $g(y) = \mathbb{E}[\phi(0^n, y)]$ (expectation over the internal randomisation). Under $p = 0$, $X = 0^n$ almost surely, so $\mathbb{P}_{0,Q}(\phi = 1) = \mathbb{E}_Q[g(Y)]$. Under $p > \varepsilon$, the event $\{X = 0^n\}$ has probability $(1 - p)^n$ and is independent of Y, hence $\mathbb{P}_{p,Q}(\phi = 1) \ge (1 - p)^n \, \mathbb{E}_Q[g(Y)] = (1 - p)^n \mathbb{P}_{0,Q}(\phi = 1)$. Validity gives $(1 - p)^n \mathbb{P}_{0,Q}(\phi = 1) \le \delta$ for all $p \in (\varepsilon, \varepsilon + \eta)$; letting $p \downarrow \varepsilon$ proves the inequality. With $\mathbb{P}_{0,Q}(\phi = 1) = 1$ it reads $(1 - \varepsilon)^n \le \delta$, i.e. $n \ge n_0$; with $\mathbb{P}_{0,Q}(\phi = 1) \ge 1 - \beta$ it gives the power bound. □

Remark 1. The argument only uses that the simulator data have the same law under the two hypotheses and that the all-success real sample has positive probability under both. It is the success-run analogue of the observation that external data cannot increase power under strict type-I error control over all external-data laws [2]. *Check.* The zero-failure test with $n = 59$ at $p = \varepsilon = 5\%$: $(0.95)^{59} = 0.0485$ (MC 0.0491, 10^5 runs).

S1.3 C2: the price of borrowing

Suppose one accepts a two-tier guarantee: level δ when the simulator is "right" and only level $\delta' > \delta$ in general.

Theorem S2 (C2, exact credit)**.** *(i)* Converse. *Under the assumption of Theorem S1, any procedure valid at level δ' on $\mathcal{P}$ that certifies a perfect system with probability one uses $n \ge n_0(\varepsilon, \delta')$ real events. Hence the credit, the number of real events saved with respect to $n_0(\varepsilon, \delta)$, is at most*

$$c = n_0(\varepsilon, \delta) - n_0(\varepsilon, \delta').$$

(ii) Attainment. *Let $T(Y) \in \{0, 1\}$ be any simulator test ("the simulator passes") and let B be the procedure: if $T = 1$ run the zero-failure test with $n_1 = n_0(\varepsilon, \delta')$ real events, otherwise with $n_0 = n_0(\varepsilon, \delta)$. Then B is valid at level δ' on $\mathcal{P}$, saves exactly c events whenever $T = 1$, and is valid at level δ on the subclass*

$$\mathcal{P}_0 = \Big\{(p, Q) : \; p > \varepsilon \Rightarrow \mathbb{P}_Q(T = 1) \le \pi^*\Big\}, \qquad \pi^* = \frac{\delta - (1 - \varepsilon)^{n_0}}{(1 - \varepsilon)^{n_1} - (1 - \varepsilon)^{n_0}},$$

i.e. where a simulator of a system that is truly worse than ε rarely passes. (iii) Continuous lower bound. $c \ge c^* = \big\lfloor \ln(\delta'/\delta)/(-\ln(1 - \varepsilon)) \big\rfloor$.

Proof. (i) Theorem S1 at level δ' gives $(1 - \varepsilon)^n \le \delta'$, whose smallest integer solution is $n_0(\varepsilon, \delta')$. (ii) For $p > \varepsilon$, with $\pi = \mathbb{P}_Q(T = 1)$ and T independent of the real data, $\mathbb{P}(B = 1) = \pi(1 - p)^{n_1} + (1 - \pi)(1 - p)^{n_0} \le (1 - \varepsilon)^{n_1} \le \delta'$, and $\le \delta$ as soon as $\pi \le \pi^*$ (the expression is affine in π and decreasing in p). (iii) Write $a = \ln \delta / \ln(1 - \varepsilon)$, $a' = \ln \delta' / \ln(1 - \varepsilon)$ and $r = a - a' = \ln(\delta'/\delta)/(-\ln(1 - \varepsilon))$. Since $n_0(\delta) \ge a$ and $n_0(\delta') < a' + 1$, $c > r - 1$; c being an integer, $c \ge \lfloor r \rfloor = c^*$. □

Remark 2 (numbers and checks). At $\varepsilon = \delta = 5\%$: $\delta' = 7.5, 10, 15, 20\%$ give $c = 8, 14, 22, 27$ ($-14\%, -24\%, -37\%, -46\%$ of 59). The exact form is maximal on all 40 grid points ($(1-\varepsilon)^{n_0(\delta')} \le \delta' < (1-\varepsilon)^{n_0(\delta')-1}$: yes); MC ($10^5$ runs per point) matches the closed form within 0.29 percentage points. *Empirical C2* (procedure B, T = the one-sided CP bound of the simulator's miss rate $\le \varepsilon$, 10^5 runs, $\varepsilon = 5\%$, $d_{\min} = 32$, $K = 1$): with the paired twin of the 48 static scenes (114 events) the simulator passes in 73.3% of runs; realised credit 5.9/10.3/16.1/19.8 events and false certification at the worst case $p = \varepsilon$ equal to 6.7%/8.4%/12.2%/15.6% for $\delta' = 7.5/10/15/20\%$ — the credit is bought exactly with the extra error $\delta' - \delta$. The off-the-shelf synthetic benchmark (30 events of length ≥ 32) never passes (0.0%): too few synthetic events to bound its own miss rate below 5%.

S1.4 C3: the twin lemma

Lemma S3 (C3)**.** *For any pairing (M, M') of a real event and its twin, $p \le q + D$ with $D = \mathbb{P}(M = 1, M' = 0)$. No assumption links the twin to reality.*

Proof. $p = \mathbb{P}(M = 1, M' = 1) + \mathbb{P}(M = 1, M' = 0) \le \mathbb{P}(M' = 1) + D = q + D$. □

Corollary S4 (a per-camera twin does not save events)**.** *If $q \le q_U$ holds with confidence $1 - \delta/2$ and $D \le \varepsilon - q_U$ is certified for the same camera by a zero-discordance success run at level $\delta/2$, the number of paired real events is $\lceil \ln(\delta/2)/\ln(1-(\varepsilon - q_U)) \rceil = 72, 91, 122$ for $q_U = 0, 1, 2\%$ at $\varepsilon = \delta = 5\%$, all larger than $n_0 = 59$. The twin can only help when D is calibrated once, on many scenes, and transferred (C4).*

Remark 3 (event unit). With a uniform phase, $x_i = \mathbb{P}_\varphi(M_i = 1, M_i' = 0) = |R_i' \setminus R_i|/K$, where R_i, R_i' are the residue sets of the real and twin hit frames of event i (twin catch decided by the majority of three sprite variants). The *worst-phase* indicator $W_i = \mathbf{1}\{R_i' \setminus R_i \ne \emptyset\}$ satisfies $\mathbb{P}(M_i = 1, M_i' = 0 \mid \text{event}) = x_i \le W_i$. All certificates below are stated for Bernoulli indicators Z_i; they hold verbatim with $Z_i = W_i$ and then bound the (larger) worst-phase discordance rate. This integer form is the one used for the reported certificates.

S1.5 fleet bound: event-weighted certificate for the calibrated fleet

Lemma S5 (mean–median)**.** *For $N \ge 2$ and $0 < p < 1 - 1/N$, $F_{N,p}(\lfloor Np \rfloor) > 1/4$. Consequently, if $\delta \le 1/4$ and $F_{N,p}(x) < \delta$, then $x \le Np - 1$.*

Proof. Greenberg and Mohri [3] prove $\mathbb{P}(B \ge \mathbb{E}B) > 1/4$ for $B \sim \mathrm{Bin}(N, r)$ with $r > 1/N$. Apply it to $B = N - \mathrm{Bin}(N, p) \sim \mathrm{Bin}(N, 1-p)$ with $1 - p > 1/N$: $\mathbb{P}(\mathrm{Bin}(N,p) \le Np) > 1/4$, and $\{\mathrm{Bin}(N,p) \le Np\} = \{\mathrm{Bin}(N,p) \le \lfloor Np \rfloor\}$. For the consequence: $F_{N,p}$ is non-decreasing, so $F_{N,p}(x) < \delta \le 1/4 < F_{N,p}(\lfloor Np \rfloor)$ forces $x < \lfloor Np \rfloor$, i.e. $x \le \lfloor Np \rfloor - 1 \le Np - 1$. *Check:* $\inf F_{N,p}(\lfloor Np \rfloor)$ over $N = 2..2000$ and a fine grid of p is 0.2500, approached at $N = 2$, $p \to 1/2$ (the constant 1/4 is tight). □

Theorem S6 (fleet bound)**.** *Let $Z_1, \dots, Z_N$ be independent, $Z_i \sim \mathrm{Bern}(d_i)$ with arbitrary $d_i \in [0,1]$ (heterogeneous across scenes and events), $X = \sum_i Z_i$ and $\bar{d} = \frac{1}{N}\sum_i d_i$. If $\delta \le 1/4$ and $\bar{d} < 1 - 1/N$, then*

$$\mathbb{P}\big(U(X, N) < \bar{d}\big) \le \delta.$$

With $d_i = D_{s(i)}$ (the discordance of the scene of event i), $\bar{d} = \bar{D}_w = \sum_s n_s D_s / N$, the event-weighted discordance of the calibrated cameras; combined with Lemma S3, $\bar{p}_w \le \bar{q}_w + U(X, N)$ with probability at least $1 - \delta$.

Proof. If $\bar{d} = 0$ there is nothing to prove. Otherwise let $x^* = \max\{x : F_{N,\bar{d}}(x) < \delta\}$ (if the set is empty the failure probability is 0). By monotonicity of U, $\{U(X, N) < \bar{d}\} = \{X \le x^*\}$. By Lemma S5, $x^* \le N\bar{d} - 1$. Hoeffding's Theorem 4 [4]: for a sum S of independent Bernoulli variables with mean $N\bar{d}$ and $0 \le b \le N\bar{d} - 1$, $\mathbb{P}(S \le b) \le F_{N,\bar{d}}(b)$. With $b = x^*$, $\mathbb{P}(X \le x^*) \le F_{N,\bar{d}}(x^*) < \delta$. □

Remark 4 (what it does not say; checks). The guarantee is conditional on the calibrated scenes and weighted by events: it does not bound any single camera and does not transfer to a new camera (if scenes are random, X is over-dispersed; e.g. $D_s \in \{0, 1\}$ gives $\mathbb{P}(X = 0) = (1 - \mu)^m \gg (1 - \mu)^N$). *Checks.* Six synthetic heterogeneity patterns on a fixed configuration of 44 scenes with $N = 304$ events (a calibration set from an earlier stage of the study, used only as a test bed for the theorem): exact Poisson–binomial failure ≤ 0.0471, MC (10^5) ≤ 0.0473. Measured discordance probabilities of the study (event level, main domain, five operating points, as measured and rescaled to mean 2% and 5%; 12 cases): exact failure ≤ 0.0272, MC ≤ 0.0267, all $\le \delta = 0.05$.

S1.6 population bound: transfer to a new camera

Proposition S7 (population bound)**.** *Let scenes $S_1, \dots, S_m$ be i.i.d. from a scene population, each with $n_s \ge 1$ paired events whose discordance indicators are independent* $\mathrm{Bern}(D_{S_s})$ *given the scene, $\mu = \mathbb{E}[D_S]$, and $J = \#\{s : X_s \ge 1\}$ with X_s the number of discordant events in scene s. Then*

$$\mathbb{P}\big(U(J, m) < \mu\big) \le \delta.$$

For a new camera drawn from the same population, $\mathbb{E}[p_{\mathrm{new}}] \le \mathbb{E}[q_{\mathrm{new}}] + U(J, m)$ with probability $\ge 1 - \delta$ over the calibration, and $\mathbb{P}(D_{\mathrm{new}} > t) \le U(J, m)/t$ (Markov).

Proof. Let $B_s = \mathbf{1}\{X_s \ge 1\}$. Given the scene, $\mathbb{P}(B_s = 1 \mid S_s) = 1 - (1 - D_{S_s})^{n_s} \ge D_{S_s}$ because $(1 - D)^n \le 1 - D$ for $n \ge 1$. Hence $\beta_s := \mathbb{P}(B_s = 1) \ge \mu$, and the B_s are independent across scenes. By a coupling ($B_s = \mathbf{1}\{V_s \le \beta_s\} \ge \mathbf{1}\{V_s \le \mu\}$ with i.i.d. uniforms V_s), J stochastically dominates $\mathrm{Bin}(m, \mu)$. Since $U(\cdot, m)$ is non-decreasing, $\mathbb{P}(U(J, m) < \mu) \le \mathbb{P}(U(\mathrm{Bin}(m, \mu), m) < \mu) \le \delta$ by Clopper–Pearson validity. The corollary follows from Lemma S3 applied to the new camera and averaged over the population. □

Remark 5 (price and checks). The resolution is set by the number of scenes: with $J = 0$, $U \le 10\%$ needs $m \ge 29$ and $U \le 5\%$ needs $m \ge 59$. *Checks.* The tight case $D_S \in \{0, 1\}$, $n_s = 1$, $m = 48$ (then $J \sim \mathrm{Bin}(m, \mu)$ exactly), $\mu \in \{1, 2, 5, 10, 20\}\%$, 10^5 runs: maximum failure 0.0413. Leave-one-scene-out on the 48 static scenes at the primary operating point ($d_{\min} = 32$, $K = 16$): $D_s > U(J_{-s}, m - 1)$ for 0 of 48 scenes (the proposition bounds the population mean, not each scene).

S1.7 per-camera form: a per-camera (quantile) certificate and its data requirement

Proposition S8 (per-camera form)**.** *Under the assumptions of Proposition S7 with $n_s \ge n_{\min}$ for all s, let $t \in (0, 1)$, $\pi_t = \mathbb{P}(D_S > t)$ and $h_t = 1 - (1 - t)^{n_{\min}}$. Then*

$$\mathbb{P}\big(\pi_t > U(J, m)/h_t\big) \le \delta.$$

Hence, with $\hat{\gamma} = U(J, m)/h_t$, a new camera satisfies $p_{\mathrm{new}} \le q_{\mathrm{new}} + t$ except on a set of cameras of population mass at most $\hat{\gamma}$, with confidence $1 - \delta$.

Proof. $\mathbb{P}(B_s = 1) \ge \mathbb{P}(D_S > t)\big(1 - (1 - t)^{n_s}\big) \ge \pi_t h_t$. The coupling argument of Proposition S7 gives $J \succeq_{\mathrm{st}} \mathrm{Bin}(m, \pi_t h_t)$, so $\mathbb{P}(U(J, m) < \pi_t h_t) \le \delta$, which is the claim. □

Remark 6 (data requirement; checks). With $J = 0$, $\hat{\gamma} \le \gamma$ requires $m \ge m_{\min}$ with $1 - \delta^{1/m_{\min}} \le \gamma$ ($m_{\min} = 59$ for $\gamma = 5\%$, 29 for $\gamma = 10\%$) and $n_{\min} \ge \ln\big(1 - U(0,m)/\gamma\big)/\ln(1-t)$ events in every scene; the total $m\,n_{\min}$ decreases to the floor $\approx \ln(1/\delta)/(\gamma t)$ as m grows. *Check:* two-point populations $D_S \in \{0, t'\}$, $t' > t$, $m = 48$, 10^5 runs: maximum failure 0.0442.

S1.8 C5a: a pessimistic-length twin is conservative under periodic sampling

Proposition S9 (C5a)**.** *Let an event occupy frames $F = \{o, \dots, o + d - 1\}$ and let the detector outcome on each frame be fixed (a deterministic function of the frame). For the truncated event $F_L = \{o, \dots, o + L - 1\}$, $L \le d$, with hit set $H_L = H \cap F_L$, the miss probability under a uniform phase, $m(L) = 1 - |H_L \bmod K|/K$, is non-increasing in L. Consequently a twin built with a pessimistic (shorter) length L has $q_L \ge q$ and discordance $D_L \le D$, so the C3 bound $q_L + D_L$ remains valid and the miss term is conservative.*

Proof. $L \le L'$ implies $H_L \subseteq H_{L'}$, hence $H_L \bmod K \subseteq H_{L'} \bmod K$ and $m(L) \ge m(L')$. A twin catch at phase φ for the truncated twin implies a catch for the full twin, so $\{M = 1, M'_L = 0\} \subseteq \{M = 1, M' = 0\}$ and $D_L \le D$; Lemma S3 holds for any pairing. □

Remark 7 (check). On the twin of the 48 static scenes, shortening every event by $x \in \{10, 25, 50\}\%$ gives non-decreasing q for every event and every $K \in \{1, 4, 8, 16, 48\}$ (all monotone: yes); e.g. at $d_{\min} = 32$, $K = 16$, q rises from 1.37% to 2.36% at $x = 50\%$.

S2 Units and event counts

Unit. Every rate is per *event* (a maximal run of frames containing the ground-truth object whose largest instance has at least 256 pixels; events shorter than $d_{\min}$ frames are dropped). Each event has three twin renderings (sprite variants); the twin catches the event at a sampling phase if at least two variants catch it. Under a uniform phase the discordance probability of an event is $x_i = |R'_i \setminus R_i|/K$; the certificates use the integer worst-phase count $W_i = \mathbf{1}\{R'_i \setminus R_i \ne \emptyset\} \ge x_i$ (remark on the event unit in Section S1), and the phase-averaged count is reported in parentheses. Table S2 shows that the variant rule (1, 2 or 3 of 3) does not change any static-domain count. Table S1 reconciles the event counts of the initial twin run with the analysed set.

S3 Full operating-point grid

Table S3 lists all 70 operating points at $\varepsilon = 20\%$. The twin route certifies a camera with no real event when $q_{\mathrm{U}} + U_{\mathrm{pop}} \le \varepsilon$ with δ split in half, U_{pop} computed without the camera. "As built" uses the twin's own miss rate bounded from three runs per event; "converged" treats q as known. Direct certification needs n_{dir} events at 80% power.

S4 Audit details

Table S7 lists the scenes whose discordance exceeds the leave-one-scene-out bound. Table S6 gives must-fail scenario 1 (temporal) and Tables S4 and S5 must-fail scenario 2 (night scenes leaked into the calibration set), including the per-camera out-of-domain pass-through; Table S14 judges the camera-jitter and night cameras with a bound from the static scenes under the worst-phase convention. The static domain is defined by the category labels of the datasets (CDnet without *PTZ*, *cameraJitter*, *nightVideos* and *turbulence*; LASIESTA without SM and MC); this definition was fixed after the initial version of this study, and Table S16 compares the static domain of the three versions. The

camera-jitter tier holds the CDnet *cameraJitter* scenes and the LASIESTA scenes with simulated camera motion (SM); its only discordant events and its only false acceptance are in *traffic*, a CDnet scene with real shake. No static scene has 59 or more qualifying events, so the "truth" of a camera is its own finite event pool (exact over phases).

S5 Sensitivity

Table S8 compares the synthetic-versus-real classifier on all target sets (main text: 0.64, 0.94, 0.77) and on CDnet targets only (0.61, 0.96, 0.75). Table S9 gives twin and real hit rates on the same frames by sprite aspect error; the within-event paired difference on clamped frames is 3.0 points over 631 event–variant pairs. Table S10 checks C5a and Table S11 the empirical C2 procedure with every simulator.

S6 Twin construction and licences

Construction. Per scene, the background plate is the per-pixel median of up to 60 frames without ground-truth foreground (longest empty run preferred). Per event and variant (seeds 42, 43, 44): the real object (ground-truth mask, dilated 3 px) is replaced by the plate; the sprite height equals the ground-truth box height, its width is the sprite aspect times the height clamped to $\pm 10\%$ of the box width, bottom-centred on the box; brightness gain is the frame luminance over a reference (clipped to $[0.35, 1.2]$) with a 5% blend of the frame mean colour; a soft elliptical shadow and a 2-px feather are applied. Person sprites follow a four-frame walking cycle, mirrored to the motion direction; vehicle sprites are chosen by aspect ratio. The detector is run on every twin frame until the hit residues cover all 48 classes modulo 48 (exact early stop for every K dividing 48).

Licences. Vehicle sprites were cut from Virtual KITTI 2 (Cabon et al., 2020; CC BY-NC-SA 3.0) and are not redistributed; the extraction script is provided. Person sprites are generated with MPFB 2 from its CC0 bundled assets. No frame of CDnet 2014 or LASIESTA is reproduced in the paper or this supplement, because no reuse terms for figures could be recorded; the illustration of sprites uses CC0 assets on a neutral background.

S7 Supplementary tables

Table S1: Event counts from the initial twin run to the analysed set (primary definition: largest object ≥ 256 px). The two headline counts differ in unit: all durations vs. duration ≥ 32.

step	events	scenes
initial twin run, original definition (all events, all durations)	329	64
initial twin run, largest object ≥ 256 px	257	64
same scenes, LASIESTA frames in true temporal order	257	64
LASIESTA scene list rebuilt (20 to 48 scenes)	308	92
pan-tilt-zoom scenes excluded (4)	283	88
LASIESTA moving-camera scenes excluded (4)	279	84
scenes run (2 held-out turbulence scenes not run)	269	82
events of at least $d_{\min} = 32$ frames	203	82

Table S2: Sprite-variant rule (twin catches if $\geq r$ of 3 variants catch) at OP*: worst-phase discordant events; phase-averaged count under the rule (exp.) and averaged over the three variants (var. mean, the parenthesised count of the main text); bounds.

tier	r	N	disc.	exp.	var. mean	$U(X,N)$	J/m	$U(J,m)$
main	1/3	114	0	0.00	0.00	2.59%	0/48	6.05%
main	2/3	114	0	0.00	0.00	2.59%	0/48	6.05%
main	3/3	114	0	0.00	0.00	2.59%	0/48	6.05%
jitter	1/3	62	3	0.31	0.31	12.03%	1/26	16.98%
jitter	2/3	62	3	0.31	0.31	12.03%	1/26	16.98%
jitter	3/3	62	3	0.31	0.31	12.03%	1/26	16.98%
night	1/3	25	13	8.75	8.21	69.49%	5/6	99.15%
night	2/3	25	12	8.31	8.21	65.86%	5/6	99.15%
night	3/3	25	12	7.56	8.21	65.86%	5/6	99.15%
turbulence	1/3	2	1	0.12	0.12	97.47%	1/2	97.47%
turbulence	2/3	2	1	0.12	0.12	97.47%	1/2	97.47%
turbulence	3/3	2	1	0.12	0.12	97.47%	1/2	97.47%
all tiers	1/3	203	17	9.19	8.65	12.30%	7/82	15.44%
all tiers	2/3	203	16	8.75	8.65	11.73%	7/82	15.44%
all tiers	3/3	203	16	8.00	8.65	11.73%	7/82	15.44%

Table S3: Full operating-point grid at $\varepsilon = 20\%$ (48 static cameras): static-domain real miss rate, events for direct certification, median events saved per camera (twin run to convergence), cameras saving (converged / as built), CPU-hours per saved event.

$d_{\min}$	K	$\hat{p}$	n_{dir}	saved	cams	as built	CPU-h
32	1	0.88%	14	14	47	3	0.064
32	2	0.88%	14	14	47	3	0.064
32	3	0.88%	14	14	47	3	0.064
32	4	0.88%	14	14	47	3	0.064
32	6	0.88%	14	14	47	3	0.064
32	8	0.88%	14	14	47	3	0.064
32	12	0.88%	14	14	47	3	0.070
32	16	0.88%	14	14	46	3	0.069
32	24	1.21%	14	14	44	0	0.351
32	48	5.94%	37	0	0	0	–
24	1	0.84%	14	14	47	3	0.064
24	2	0.84%	14	14	47	3	0.064
24	3	0.84%	14	14	47	3	0.064
24	4	0.84%	14	14	47	3	0.064
24	6	0.84%	14	14	47	3	0.064
24	8	1.05%	14	14	47	0	0.094
24	12	1.12%	14	14	45	0	0.160
24	16	1.68%	22	22	45	0	0.106
24	24	2.28%	22	22	43	0	0.229
24	48	8.23%	57	0	0	0	–
16	1	1.55%	14	14	47	4	0.064
16	2	1.94%	22	22	47	3	0.041
16	3	2.07%	22	22	47	3	0.041
16	4	2.13%	22	22	47	3	0.043
16	6	2.20%	22	22	46	3	0.042
16	8	2.62%	22	22	46	0	0.061
16	12	3.42%	22	22	45	0	0.102
16	16	4.26%	30	30	45	0	0.078

$d_{\min}$	K	$\hat{p}$	n_{dir}	saved	cams	as built	CPU-h
16	24	6.07%	37	37	43	0	0.136
16	48	13.45%	210	0	0	0	–
12	1	2.31%	22	22	47	4	0.041
12	2	2.69%	22	22	47	3	0.041
12	3	2.82%	22	22	47	3	0.041
12	4	2.88%	22	22	47	3	0.043
12	6	2.95%	22	22	46	3	0.042
12	8	3.37%	22	22	46	0	0.061
12	12	4.17%	30	30	45	0	0.074
12	16	5.00%	30	30	45	0	0.078
12	24	6.79%	44	44	43	0	0.115
12	48	14.12%	260	0	0	0	–
8	1	2.22%	22	22	47	4	0.041
8	2	2.59%	22	22	47	4	0.041
8	3	2.72%	22	22	47	3	0.043
8	4	2.96%	22	22	47	0	0.063
8	6	2.96%	22	22	46	3	0.042
8	8	3.98%	30	30	46	0	0.073
8	12	5.56%	37	37	44	0	0.062
8	16	6.90%	44	44	44	0	0.053
8	24	9.17%	69	69	43	0	0.073
8	48	16.76%	901	0	0	0	–
4	1	2.72%	22	22	47	4	0.041
4	2	3.40%	22	22	47	3	0.043
4	3	4.08%	30	30	46	3	0.031
4	4	4.76%	30	30	46	0	0.045
4	6	6.35%	44	44	46	0	0.031
4	8	8.42%	63	63	45	0	0.035
4	12	11.00%	106	106	44	0	0.021
4	16	12.80%	170	170	44	0	0.014
4	24	15.45%	441	441	43	0	0.011
4	48	22.99%	–	0	0	0	–
1	1	3.11%	22	22	47	0	0.063
1	2	5.90%	37	37	46	0	0.059
1	3	7.66%	50	50	46	0	0.044
1	4	9.47%	76	76	45	0	0.029
1	6	12.11%	135	135	45	0	0.017
1	8	14.60%	310	310	45	0	0.007
1	12	17.55%	1583	1583	44	0	0.001
1	16	19.49%	–	0	0	0	–
1	24	22.20%	–	0	0	0	–
1	48	29.39%	–	0	0	0	–

Table S4: Night scenes at OP*: real miss rate split into detector and temporal misses, twin miss rate, discordance, and whether the camera is falsely certified when judged with a bound from the static scenes only.

scene	n	p_s	detector	temporal	q_s	D_s	disc.	false cert.
bridgeEntry	2	15.6%	15.6%	0.0%	0.0%	15.6%	1	no
busyBoulvard	4	96.9%	96.9%	0.0%	0.0%	96.9%	4	yes
fluidHighway	4	26.6%	26.6%	0.0%	0.0%	26.6%	2	yes
streetCornerAtNight	10	28.1%	28.1%	0.0%	11.9%	20.6%	4	yes
tramStation	2	0.0%	0.0%	0.0%	0.0%	0.0%	0	no
winterStreet	3	33.3%	33.3%	0.0%	0.0%	33.3%	1	yes

Table S5: Must-fail 2 at OP*: k night scenes in the calibration set (all subsets).

k	subsets	U_{pop} med.	range	U_{fleet} med.	withdrawn	static false cert.
0	1	6.1%	6.1%–6.1%	2.6%	0%	0
1	6	9.3%	5.9%–9.3%	4.6%	0%	0
2	15	12.1%	9.1%–12.1%	7.5%	67%	0
3	20	13.2%	11.8%–14.5%	9.1%	100%	0
4	15	14.2%	14.2%–16.7%	10.6%	100%	0
5	6	16.4%	16.4%–18.8%	12.6%	100%	0
6	1	18.5%	18.5%–18.5%	13.6%	100%	0

Table S6: Must-fail 1 (temporal, $K = 48$): static scenes with $p_s > \varepsilon$; detector vs. temporal misses and twin miss rate. All were refused in every Monte-Carlo run.

point	scene	n	p_s	detector	temporal	q_s
d1 K48	port_0_17fps	5	99.6%	43.3%	56.2%	100.0%
d1 K48	tramCrossroad_1fps	11	77.7%	6.1%	71.6%	74.2%
d1 K48	tunnelExit_0_35fps	34	83.6%	5.2%	78.4%	87.9%
d4 K48	port_0_17fps	5	99.6%	43.3%	56.2%	100.0%
d4 K48	tramCrossroad_1fps	8	71.1%	8.1%	63.0%	66.7%
d4 K48	tunnelExit_0_35fps	23	77.3%	7.5%	69.7%	83.4%
d16 K48	port_0_17fps	3	99.3%	57.6%	41.7%	100.0%
d16 K48	tramCrossroad_1fps	4	55.7%	12.5%	43.2%	47.9%
d16 K48	tunnelExit_0_35fps	11	61.7%	11.9%	49.8%	72.7%

Table S7: Leave-one-scene-out: scenes whose discordance exceeds the bound computed without them.

point	scene	n	D_s	p_s	q_s	U_{-s}	$p_s > q_s + U$	false cert.
–	none							

Table S8: Synthetic-vs-real classifier (logistic), median AUC over target sets: all targets (main text) and CDnet targets only.

features	all targets	CDnet targets
temporal	0.64	0.61
geometry	0.94	0.96
appearance only	0.77	0.75

Table S9: Twin and real hit rates on the same frames by object kind and sprite aspect error (before the width clamp).

kind	|aspect err.|	frames	twin hit	real hit	diff. (pts)
person	≤10%	8870	91.7%	94.8%	-3.2
person	10-20%	5899	90.1%	94.2%	-4.2
person	20-40%	4404	74.0%	86.2%	-12.2
person	>40%	5449	34.4%	44.5%	-10.0
vehicle	≤10%	4105	86.9%	89.0%	-2.1
vehicle	10-20%	2569	88.0%	90.2%	-2.2
vehicle	20-40%	3087	87.6%	85.1%	2.5
vehicle	>40%	3881	64.9%	74.2%	-9.3

Table S10: C5a: twin miss rate when every event is shortened by x.

$d_{\min}$	K	$x = 0$	10%	25%	50%	per event monotone
32	1	0.88%	0.88%	0.88%	0.88%	yes
32	8	0.99%	0.99%	0.99%	1.21%	yes
32	16	1.37%	1.43%	1.59%	2.36%	yes
32	48	7.05%	7.80%	10.29%	19.85%	yes
16	1	2.33%	2.33%	2.33%	2.33%	yes
16	8	4.36%	4.46%	4.65%	5.72%	yes
16	16	6.54%	6.73%	7.17%	9.16%	yes
16	48	15.04%	15.94%	18.56%	27.65%	yes
1	1	4.97%	4.97%	5.59%	6.83%	yes
1	8	16.69%	16.85%	18.01%	21.04%	yes
1	16	21.66%	21.89%	22.83%	25.50%	yes
1	48	30.77%	31.52%	33.81%	41.46%	yes

Table S11: Empirical C2: two-tier procedure with each simulator (100,000 runs), $\delta' = 7.5/10/15/20\%$.

point	simulator	N_{sim}	pass	credit (events)	false cert. at $p = \varepsilon$
$\varepsilon = 5\%$, (32,1)	BMC (G*)	30	0.0%	0.0/0.0/0.0/0.0	4.9%/4.8%/4.8%/4.8%
$\varepsilon = 5\%$, (32,1)	BMC (temporal)	49	0.0%	0.0/0.0/0.0/0.0	4.8%/4.8%/4.7%/4.9%
$\varepsilon = 5\%$, (32,1)	paired twin	114	73.3%	5.9/10.3/16.1/19.8	6.7%/8.4%/12.2%/15.6%
$\varepsilon = 20\%$, OP*	BMC (G*)	30	100.0%	2.0/3.0/5.0/6.0	7.0%/8.5%/13.4%/16.7%
$\varepsilon = 20\%$, OP*	BMC (temporal)	49	100.0%	2.0/3.0/5.0/6.0	6.9%/8.6%/13.4%/17.0%
$\varepsilon = 20\%$, OP*	paired twin	114	100.0%	2.0/3.0/5.0/6.0	6.8%/8.5%/13.5%/16.8%

Table S12: Worst-phase convention (valid for any fixed deployment phase): real and twin miss rates averaged over phases and at the worst phase, discordant events, fleet and population bounds.

point	tier	N	$\hat{p}$ avg	$\hat{p}$ worst	$\hat{q}$ avg	$\hat{q}$ worst	disc.	$U(X,N)$	J/m	$U(J,m)$
OP*	main	114	0.9%	0.9%	1.4%	3.5%	0	2.59%	0/48	6.05%
	jitter	62	0.5%	4.8%	0.0%	0.0%	3	12.03%	1/26	16.98%
	night	25	36.2%	52.0%	4.8%	16.0%	12	65.86%	5/6	99.15%
	turbulence	2	6.2%	50.0%	0.0%	0.0%	1	97.47%	1/2	97.47%
	all tiers	203	5.2%	8.9%	1.4%	3.9%	16	11.73%	7/82	15.44%

Table S13: Exchange rate under the worst-phase convention (48 static cameras; twin run to convergence).

point	$\hat{p}$ worst	n_{dir}	accepted	saved	false acc.	twin runs
OP*	0.9%	14	44	14	0	28

Table S14: Out-of-domain cameras (camera jitter, night) judged by the twin route calibrated on the static scenes only (worst-phase convention, transfer term $U_{\delta/2}(J, m)$ of the static scenes): accepted, and falsely accepted ($p_w > \varepsilon$).

point	tier	camera	n	p_w	q_w	$U_{\delta/2}$	accepted	false acc.
OP*	jitter	I_SM_01	3	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_02	2	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_03	2	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_04	3	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_05	3	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_06	3	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_07	3	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_08	3	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_09	3	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_10	3	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_11	3	0.0%	0.0%	7.4%	yes	no
	jitter	I_SM_12	3	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_01	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_02	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_03	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_04	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_05	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_06	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_07	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_08	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_09	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_10	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_11	1	0.0%	0.0%	7.4%	yes	no
	jitter	O_SM_12	1	0.0%	0.0%	7.4%	yes	no
	jitter	boulevard	8	0.0%	0.0%	7.4%	yes	no
	jitter	traffic	8	37.5%	0.0%	7.4%	yes	yes
	night	bridgeEntry	2	50.0%	0.0%	7.4%	yes	yes
	night	busyBoulvard	4	100.0%	0.0%	7.4%	yes	yes
	night	fluidHighway	4	50.0%	0.0%	7.4%	yes	yes
	night	streetCornerAtNight	10	50.0%	40.0%	7.4%	no	no
	night	tramStation	2	0.0%	0.0%	7.4%	yes	no
	night	winterStreet	3	33.3%	0.0%	7.4%	yes	yes

Table S15: Site-level population bound at OP*: 17 sites, $J_{\text{site}} = 0$, $U(J_{\text{site}}, m_{\text{site}}) = 16.2\%$; per site: scenes, events, discordant events, bound without the site, leave-one-site-out violation.

site	scenes	events	disc.	$U_{-\text{site}}$	violation
CDnet badWeather	4	15	0	17.1%	no
CDnet baseline	1	2	0	17.1%	no
CDnet dynamicBackground	6	19	0	17.1%	no
CDnet intermittentObjectMotion	3	4	0	17.1%	no
CDnet lowFramerate	4	5	0	17.1%	no
CDnet shadow	6	37	0	17.1%	no
CDnet thermal	4	7	0	17.1%	no
LASIESTA I_BS	2	2	0	17.1%	no
LASIESTA I_CA	2	2	0	17.1%	no
LASIESTA I_IL	2	2	0	17.1%	no
LASIESTA I_MB	2	2	0	17.1%	no
LASIESTA I_OC	2	4	0	17.1%	no
LASIESTA I_SI	2	4	0	17.1%	no
LASIESTA O_CL	2	2	0	17.1%	no
LASIESTA O_RA	2	3	0	17.1%	no
LASIESTA O_SN	2	2	0	17.1%	no
LASIESTA O_SU	2	2	0	17.1%	no

Table S16: Static domain at OP* in the three versions of this study: 78 scenes; 76 scenes (CDnet camera-jitter scenes moved to the jitter tier); 48 scenes (LASIESTA simulated-motion scenes moved to the jitter tier and LASIESTA moving-camera scenes excluded; this version). Each column is recomputed by the released code from the same twin run and, for the two earlier versions, equals the numbers of those internal, unpublished drafts (tags ress-v1.0, ress-v1.1); the last column gives the scenes that carry the earlier disagreements.

quantity	78 scenes	76 scenes	48 scenes	located in
static scenes / events	78 / 180	76 / 164	48 / 114	–
discordant events (worst phase), $U(X, N)$	3, 4.25%	0, 1.81%	0, 2.59%	traffic
discordant scenes J/m, $U(J, m)$	1/78, 5.94%	0/76, 3.87%	0/48, 6.05%	traffic
scenes above their LOSO bound	1/78	0/76	0/48	traffic
worst-phase exchange: n_{dir}; accepted; false acc.	22; 74/78; 1	14; 72/76; 0	14; 44/48; 0	traffic

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